

Brane Thickness δ from 5D Action

Deriving the Proton-Scale Length from First Principles

EDC Project

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Abstract

We derive the brane thickness parameter δ from the 5D brane-world action. The key result is that for a thick brane with tension σ supporting a localized fermion (the proton), the characteristic thickness is set by $\delta = \hbar/(2m_p c)$ where m_p is the proton mass. This follows from matching the bound-state energy of the localized mode to the observed proton mass. The derivation is model-dependent [Dc] but provides a principled connection between the geometric parameter and observable physics.

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1 Definitions and Setup

1.1 What δ Means Geometrically

Definition 1.1 (Brane Thickness δ). [Dc] The brane thickness δ is the characteristic length scale over which fields transition from bulk behavior to brane-localized behavior in the extra dimension χ (or y). Operationally:

$$\delta = \text{width of the effective potential well in } \chi \quad (1)$$

Three equivalent interpretations of δ :

1. **Localization width:** For a mode profile $w(\chi)$ localized on the brane, δ is the standard deviation:

$$\delta^2 \sim \frac{\int \chi^2 |w(\chi)|^2 d\chi}{\int |w(\chi)|^2 d\chi} \quad (2)$$

2. **Potential well width:** For an effective potential $V(\chi)$ with a well near $\chi = 0$, δ is the half-width at half-maximum.
3. **Curvature radius:** For a warp factor $A(\chi)$ with $A''(0) = -1/\ell^2$, the curvature radius at the brane is $\ell \sim \delta$.

1.2 Unit Conventions

Throughout this document, we use **natural units** with $\hbar = c = 1$:

Quantity	Dimension
Length	$[\text{energy}]^{-1}$
Time	$[\text{energy}]^{-1}$
Mass	$[\text{energy}]$
δ	$[\text{energy}]^{-1}$
Brane tension σ	$[\text{energy}]^4$
5D gravitational coupling κ_5^2	$[\text{energy}]^{-3}$

To restore SI units: $\delta_{\text{SI}} = \hbar c \cdot \delta_{\text{natural}}$.

2 The 5D Action

2.1 Bulk + Brane Decomposition

Postulate 2.1 (5D Brane-World Action). [P] *The total action consists of bulk gravity, brane tension, and matter:*

$$S_{5D} = S_{\text{bulk}} + S_{\text{brane}} + S_{\text{matter}} \quad (3)$$

with:

$$S_{\text{bulk}} = \frac{1}{2\kappa_5^2} \int d^5x \sqrt{-g} (R_5 - 2\Lambda_5) \quad (4)$$

$$S_{\text{brane}} = -\sigma \int d^4\xi \sqrt{-h} \quad (5)$$

$$S_{\text{matter}} = \int d^5x \sqrt{-g} \mathcal{L}_m \quad (6)$$

where σ is the brane tension, h is the induced metric on the brane, and Λ_5 is the bulk cosmological constant.

2.2 Which Terms Set Which Scales

Proposition 2.2 (Scale Identification). *[Dc] In the 5D action:*

1. The **kinetic/gradient term** (from R_5) provides the χ -derivative penalty:

$$S_{kin} \sim \frac{1}{\kappa_5^2} \int (\partial_\chi \phi)^2 d\chi \quad (7)$$

This penalizes rapid variation in χ and tends to spread modes.

2. The **brane tension term** σ provides localization:

$$S_{loc} \sim -\sigma \int \delta(\chi) d\chi \quad (8)$$

For a thick brane, the delta function is smoothed over width δ .

The competition between spreading (gradient) and localization (tension) determines the equilibrium thickness δ .

3 Effective 1D Schrödinger Problem

3.1 Dimensional Reduction

Proposition 3.1 (Mode Equation). *[Der] After KK decomposition, the transverse mode profile $w(\chi)$ satisfies:*

$$\boxed{-\frac{d^2 w}{d\chi^2} + V(\chi)w = \lambda w} \quad (9)$$

where $V(\chi)$ is an effective potential and λ is the KK eigenvalue.

Proof. Standard KK reduction. For a 5D field $\Phi(x^\mu, \chi)$, write $\Phi = \sum_n \phi_n(x^\mu) w_n(\chi)$. The 5D Klein-Gordon or Dirac equation separates into 4D equations for ϕ_n and the transverse eigenvalue problem (9). $\square$

3.2 Potential Structure

Proposition 3.2 (Potential from Brane Tension). *[Dc] For a thick brane with tension σ smoothed over width δ :*

$$V(\chi) = -V_0 \cdot f\left(\frac{\chi}{\delta}\right) \quad (10)$$

where $f(x)$ is a normalized profile ($f(0) = 1$, $\int f dx \sim 1$) and V_0 is the potential depth.

Proposition 3.3 (Depth-Width Relation). *[Dc] Dimensional analysis constrains:*

$$V_0 \sim \frac{1}{\delta^2} \quad (11)$$

Proof. The potential V has dimension [energy]² in natural units. The only length scale available is δ . Therefore $V_0 \sim 1/\delta^2$. $\square$

4 The Proton as Bound State

4.1 Bound State Energy

Proposition 4.1 (Ground State Energy). *[Der] For a potential well of depth $V_0 \sim 1/\delta^2$ and width δ , the ground state energy (measured from the bottom of the well) is:*

$$E_0 = \frac{c_E}{\delta} \quad (12)$$

where $c_E = O(1)$ is a dimensionless coefficient that depends on the potential shape.

Proof. This is the standard quantum mechanical result. For a particle in a well, the characteristic frequency is $\omega \sim \sqrt{V''/M_{\text{eff}}}$ where $V'' \sim V_0/\delta^2 \sim 1/\delta^4$ is the curvature at the minimum. With effective mass $M_{\text{eff}} \sim 1/\delta$ (from the kinetic structure), we get:

$$\omega \sim \sqrt{\frac{1/\delta^4}{1/\delta}} = \sqrt{\frac{1}{\delta^3}} \cdot \delta^{1/2} = \frac{1}{\delta} \quad (13)$$

The ground state energy is $E_0 = c_E \cdot \omega$ where c_E encodes the precise potential shape (e.g., $c_E = 1/2$ for harmonic oscillator).

Key point: The $1/\delta$ dependence is robust (dimensional); the coefficient c_E is model-dependent. $\square$

4.2 Matching to Proton Mass

Postulate 4.2 (Proton as Localized Mode). *[P] The proton is (part of) the ground state of the localized fermionic mode on the brane. Its mass m_p is determined by the bound state energy:*

$$m_p = E_0 \quad (14)$$

Theorem 4.3 (Brane Thickness from Proton Mass). *[Dc] Under Postulate 4.2, the brane thickness is:*

$$\delta = \frac{c_\delta}{m_p} \quad (15)$$

where c_δ is a dimensionless constant of order unity.

Proof. From Proposition 4.1: $E_0 \sim 1/\delta$. From Postulate 4.2: $E_0 = m_p$. Therefore:

$$m_p \sim \frac{1}{\delta} \quad \Rightarrow \quad \delta \sim \frac{1}{m_p} \quad (16)$$

The precise coefficient c_δ depends on the potential shape. $\square$

4.3 Determining the Coefficient

Proposition 4.4 (Coefficient from Harmonic Approximation). *[Dc] For a harmonic potential $V(\chi) \approx V_{\min} + \frac{1}{2}M_{\text{eff}}\omega^2\chi^2$ near the minimum, the ground state energy is:*

$$E_0 = \frac{\omega}{2} \quad (17)$$

where $\omega = \sqrt{V''(0)/M_{\text{eff}}}$ is the oscillation frequency. Matching $E_0 = m_p$ with $\omega \sim 1/\delta$ gives:

$$c_\delta = \frac{1}{2} \quad (18)$$

Proof. The frequency ω carries all the shape information:

$$\omega = \sqrt{\frac{V''(0)}{M_{\text{eff}}}} \sim \sqrt{\frac{V_0/\delta^2}{1/\delta}} = \sqrt{\frac{1}{\delta^3}} \cdot \delta^{1/2} = \frac{\tilde{c}_\omega}{\delta} \quad (19)$$

where $\tilde{c}_\omega = O(1)$ depends on the exact curvature $V''(0)$.

For a harmonic oscillator specifically, $E_0 = \omega/2$, so:

$$m_p = E_0 = \frac{\omega}{2} = \frac{\tilde{c}_\omega}{2\delta} \quad \Rightarrow \quad \delta = \frac{\tilde{c}_\omega}{2m_p} \quad (20)$$

With $\tilde{c}_\omega \approx 1$ (calibrated to match units), we get $c_\delta = 1/2$.

Open: The exact value of c_δ depends on $V''(0)/\delta^4$. Deriving $V(\chi)$ from the bulk EOM would fix this coefficient precisely. $\square$

Main Result

$$\delta = \frac{1}{2m_p} = \frac{\hbar}{2m_p c} = 0.105 \text{ fm} = 0.533 \text{ GeV}^{-1} \quad (21)$$

In words: The brane thickness is half the proton Compton wavelength.

Mechanism: The brane supports a bound fermion mode whose mass equals the proton mass. The characteristic localization width of this mode is δ .

5 Consistency Checks

5.1 Numerical Verification

Quantity	Formula	Value
Proton mass	m_p	938.27 MeV [BL]
Proton Compton wavelength	$\lambda_p = 1/m_p$	0.210 fm [BL]
Predicted δ	$1/(2m_p)$	0.105 fm [Dc]
Predicted δ	$1/(2m_p)$	0.533 GeV ⁻¹ [Dc]

5.2 Comparison with Other Scales

Scale	Value (fm)	Ratio to δ	Status
$\delta = 1/(2m_p)$	0.105	1.00	[Dc]
Proton charge radius r_p	0.84	8.0	[BL]
QCD string tension ^{-1/2}	~ 0.2	~ 2	[BL]
Lattice QCD spacing	0.05–0.1	0.5–1	[BL]

The derived δ is **consistent** with QCD scales (same order of magnitude).

6 Assumptions and Limitations

6.1 Explicit Assumptions

1. **Thick brane structure [P]:** The brane has finite thickness, not a delta-function localization.

2. **Proton = bound mode [P]**: The proton is identified with a localized fermionic mode on the brane.
3. **Single scale dominance [Dc]**: The bound state energy is determined primarily by the potential width δ , not by other parameters.
4. **Harmonic approximation [Dc]**: The coefficient $c_\delta = 1/2$ assumes a roughly harmonic potential near the minimum.

6.2 What Is NOT Derived

- The **shape** of the potential $V(\chi)$ (Gaussian vs RS-like vs tanh).
- The **brane tension** σ in terms of fundamental parameters.
- The **proton mass** itself (this is input, not output).
- Why the **coefficient is exactly** $1/2$ (vs $1/3$ or $1/\pi$).

7 Stoplight Verdict

Epistemic Status Summary

Result	Status	Comment
Mode equation structure	[Der]	Standard KK reduction
$V_0 \sim 1/\delta^2$ scaling	[Der]	Dimensional analysis
$E_0 \sim 1/\delta$ scaling	[Der]	Quantum mechanics
Proton = bound mode	[P]	Postulate (model assumption)
$\delta = c/m_p$ relation	[Dc]	Derived IF proton = bound mode
$c_\delta = 1/2$ coefficient	[Dc]	Harmonic approximation
$\delta = 1/(2m_p) = 0.533 \text{ GeV}^{-1}$	[Dc]	Final result

Overall verdict: YELLOW [Dc]

The derivation is **principled but model-dependent**. The key postulate (proton = bound mode) is physically motivated but not proven from first principles. The coefficient $c_\delta = 1/2$ is approximate.

Upgrade path to GREEN:

1. Derive the potential shape $V(\chi)$ from the 5D action explicitly.
2. Compute the exact bound state energy (not just scaling).
3. Justify the proton identification from topology/homotopy arguments.

8 Conclusion

We have derived the brane thickness parameter:

$$\delta = \frac{\hbar}{2m_p c} = 0.533 \text{ GeV}^{-1} = 0.105 \text{ fm} \quad (22)$$

The derivation chain is:

1. 5D action $\rightarrow$ effective 1D Schrödinger problem [\[Der\]](#)
2. Potential well of width δ and depth $\sim 1/\delta^2$ [\[Dc\]](#)
3. Ground state energy $E_0 = c_E/\delta$ where $c_E = O(1)$ [\[Der\]](#)(scaling), [\[Dc\]](#)(coefficient)
4. Proton = bound mode $\Rightarrow m_p = E_0$ [\[P\]](#)
5. Therefore $\delta = c_\delta/m_p$ with $c_\delta = 1/2$ from harmonic approx [\[Dc\]](#)

This upgrades δ from a **physical prior** (heuristic identification) to a **derived quantity** (from 5D physics + proton identification).

What is robust: The scaling $\delta \propto 1/m_p$ follows from dimensional analysis given the bound-state interpretation.

What is model-dependent: The coefficient $c_\delta = 1/2$ comes from harmonic approximation with $E_0 = \omega/2$ and $\omega \sim \sqrt{V''/M_{\text{eff}}}$. The exact coefficient requires knowing $V''(0)$ from the bulk equations of motion.

Open problem: Derive the potential curvature $V''(0)$ from the 5D action to fix c_δ precisely (vs $1/2$, $1/\pi$, etc.).